

VISITOR MANAGEMENT SYSTEM USING SERVICENOW

Milestone 3: Business Logic and Security

Objective

Implement business logic, form customization, validation rules, and user interface configurations to automate visitor management processes and improve user experience.

Step 1: Configure Form Layout

Navigation:

Open Visitor Form

Right Click

Configure

Form Layout

Arrange fields in a logical order to improve usability and data entry efficiency.

The screenshot shows the ServiceNow Form Layout configuration interface for the 'Visitor_details' form. The interface is divided into several sections:

- Top Navigation:** Includes tabs for 'Visitor_details', 'Data', 'Forms', 'Flows', 'Policies and rules', and a 'Visitor Manage...' dropdown. There are also buttons for 'Preview' and 'Default view'.
- Left Panel (Add form elements):** Contains a search bar and a list of fields and components. Fields include 'Created Date/Time', 'Created by String', 'Employee Host Reference: Employee Details', 'Updated Date/Time', 'Updated by String', and 'Updates Integer'. There is also an option to 'Add a field in the table'.
- Center Panel (Form Layout):** Displays the 'Visitor_details' form layout. It is currently set to 'Two-column' layout. The form fields are arranged in a logical order: 'Visitor Name *', 'Email *', 'Phone Number' (with a dropdown for '+55'), 'Visit date *', 'Employee Host *' (with a dropdown for 'Employee Details'), 'Status' (with a dropdown for 'Requested'), 'Company Name', 'Checked In Time' (with a dropdown for '2008-12-06 1...'), 'Purpose of Visit *', and 'Checked Out Time' (with a dropdown for '2008-12-06 1...').
- Right Panel (Section):** Contains a 'Section label' dropdown set to 'Visitor_details'. Below it is a 'Section layout' section with 'One column' and 'Two columns' options. There is also a 'Merge with section above' button. At the bottom, there is a 'Learn more about sections' link and a diagram illustrating how sections can be merged.

Step 2: Create UI Policies

Purpose

Dynamically control field visibility and improve the user experience by displaying relevant fields based on the visitor's status.

Navigation

System UI → UI Policies

Create New UI Policy

UI Policy 1: Display Check In Time

Table

Visitor

Condition

Status = Checked In

OR

Status = Checked Out

UI Policy Action

- Check In Time → Visible = True
- Check In Time → Read Only = True

Purpose

Displays the Check In Time field after the visitor has been checked in and prevents users from modifying the recorded value.

The screenshot displays the configuration interface for a UI Policy named 'Show Check in time' within the 'Visitor Management System' application. The policy is active and associated with the 'Visitor_details [x_1877525_visit...]' table. The short description is 'Show Check in time'.

When to Apply

Conditions:

- AND: Status is Checked In
- OR: Status is Checked Out

Buttons: Update, Delete

Related Links

- [Advanced view](#)
- [Run Point Scan](#)
- [\[SN Utils\] Versions \(1\)](#)
- [\[SN Utils\] Versions \(1\)](#)

UI Policy Actions (1)

UI Policy Related List Actions

for text Search

Actions on selected rows... New

Field name	Mandatory	Visible	Read only
checked_in_time	Leave alone	True	True

1 to 1 of 1

UI Policy Action checked_in_time

UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More Info](#)

UI policy: Show Check in time

Application: Visitor Management System

Table: Visitor_details [x_1877525_visito_0_visitor_details]

* Field name: Checked In Time

Mandatory: Leave alone

Visible: True

Read only: True

Clear the field value: ☐

Buttons: Update, Delete

[\[SN Utils\] Versions \(1\)](#)

UI Policy 2: Display Check Out Time

Table

Visitor

Condition

Status = Checked Out

UI Policy Action

- Check Out Time → Visible = True
- Check Out Time → Read Only = True

Purpose

Displays the Check Out Time field only after the visitor has been checked out and prevents modification of the recorded timestamp.

The screenshot displays the configuration for a UI Policy named 'Show Check Out Time' within the 'Visitor Management System' application. The policy is active and its short description is 'Show Check Out Time'. The 'When to Apply' section shows a condition where 'Status' is 'Checked Out'. Below this, there are 'Update' and 'Delete' buttons, and a 'Related Links' section with links to 'Advanced view', 'Run Point Scan', and '[SN Utils] Versions (1)'. A section titled 'UI Policy Actions (1)' shows a table of actions for the policy.

Field name	Mandatory	Visible	Read only
checked_out_time	Leave alone	True	True

Below the table, there are navigation controls showing '1 to 1 of 1'.

The second part of the screenshot shows the 'UI Policy Action' configuration for 'checked_out_time'. It specifies the 'Field name' as 'Checked Out Time' and sets 'Mandatory' to 'Leave alone', 'Visible' to 'True', and 'Read only' to 'True'. There is also a 'Clear the field value' checkbox.

Step 3: Create Custom Role

Navigation:

sys_user_role.list

Role 1:

x_1877525_visito_0.admin

Assign role to Security.

Role 2:

x_1877525_visito_0.user.
Assign role To Employees.

The image shows two screenshots of a role configuration interface. The top screenshot is for the role 'x_1877525_visito_0.admin'. It has a suffix of 'admin', a name of 'x_1877525_visito_0.admin', and a description of 'Default admin role'. The application is 'Visitor Mangement System'. Below the form are 'Update' and 'Delete' buttons, and a section for '[SN Utils] Versions (1)'. The bottom screenshot is for the role 'x_1877525_visito_0.user'. It has a suffix of 'user', a name of 'x_1877525_visito_0.user', and a description of 'Default user role'. The application is 'Visitor Mangement System'. Below the form are 'Update' and 'Delete' buttons, and a section for '[SN Utils] Versions (1)'. Both screenshots show a 'Contains' section with a search bar and a list of items.

Role: x_1877525_visito_0.admin

* Suffix: admin

Name: x_1877525_visito_0.admin

Application: Visitor Mangement System

Elevated privilege: ☐

Assignable by:

Description: Default admin role

Update Delete

[SN Utils] Versions (1)

Contains Roles Applications with Role (1) Modules with Role (5) Custom Tables

for text Search

Role = x_1877525_visito_0.admin

Contains

Role: x_1877525_visito_0.user

* Suffix: user

Name: x_1877525_visito_0.user

Application: Visitor Mangement System

Elevated privilege: ☐

Assignable by:

Description: Default user role

Update Delete

[SN Utils] Versions (1)

Contains Roles Applications with Role (1) Modules with Role (5) Custom Tables

for text Search

Role = x_1877525_visito_0.user

Contains

Step 4: Create UI Actions

Purpose

Provide administrators with quick actions to manage visitor requests directly from the Visitor form.

UI Actions Created

- Approve
- Reject
- Check In
- Check Out

Configuration

Each UI Action updates the visitor status and performs the required business logic.

Purpose

These actions automate the visitor approval and visit tracking process, reducing manual effort and improving operational efficiency.

UI Action

Approve

Name

Approve

Table

Visitor_details [x_1877525_visit... ▼]

Order

100

Action name

Approve

Active

☒

Show insert

☒

Show update

☒

Client

☐

Overrides

Application

Visitor Mangement System

Form button

☒

Form context menu

☐

Form link

☐

Form style

-- None -- ▼

Form button icon

Show form icon only

☐

List banner button

☐

List bottom button

☐

List context menu

☐

List choice

☐

List link

☐

UI Action

Reject

Name

Reject

Table

Visitor_details [x_1877525_visit... ▼]

Order

100

Action name

Reject

Active

☒

Show insert

☒

Show update

☒

Client

☐

Overrides

Q

Application

Visitor Management System

0

Form button

☒

Form context menu

☐

Form link

☐

Form style

-- None -- ▼

Form button icon

Show form icon only

☐

List banner button

☐

List bottom button

☐

List context menu

☐

Update

Delete

UI Action
Checked In

Name

Table

Order

Action name

Active ☒

Show insert ☒

Show update ☒

Client ☐

Overrides

Application ⓘ

Form button ☒

Form context menu ☐

Form link ☐

Form style

Form button icon

Show form icon only ☐

List banner button ☐

List bottom button ☐

List context menu ☐

List choice ☐

Comments

Hint

Condition

Script ☐ Turn on ECMAScript 2021 (ES12) mode ⓘ

```

1 var now = new GlideDateTime();
2 var visitDateTime = new GlideDateTime(current.visit_date);
3
4 if (now.before(visitDateTime)) {
5     gs.addErrorMessage("Visitor cannot check in before the scheduled visit time.");
6     action.setRedirectURL(current);
7 }
8 else{
9     current.status = 'checked_in';
10
11     var ciDate = new GlideDateTime();
12     current.checked_in_time = ciDate.getValue();
13
14     current.update();
15
16     gs.addInfoMessage("Visitor Checked In Successfully");
17     action.setRedirectURL(current);
            
```

Protection policy

- Client Script

Validdate Visit date

Name

Validdate Visit date

Table

Visitor_details [x_1877525_visit...

UI Type

All

Type

onSubmit

Application

Visitor Management System

Active

☒

Inherited

☐

Global

☒

Description

Messages

Script

```

1 function onSubmit() {
2
3     var visitDate = g_form.getValue('visit_date');
4
5     if (!visitDate)
6         return true;
7
8     var selectedDate = new Date(visitDate);
9     var now = new Date();
10
11     // Past date/time validation
12     if (selectedDate < now) {
13         g_form.addErrorMessage("Visit date and time cannot be in the past.");
14         return false;
15     }
16
17     // Working hours validation (9 AM - 6 PM)
18     var hour = selectedDate.getHours();
19

```

Learning Outcomes

- Learned to create and configure UI Policies.
- Implemented dynamic field visibility and read-only controls.
- Created Client Scripts for date and time validation.
- Applied business logic to enforce working-hour restrictions.
- Developed UI Actions for visitor approval, rejection, check-in, and check-out processes.
- Improved data quality through client-side validation.
- Enhanced user experience by automating visitor management workflows.